Concrete Conflict-Analysis Calculus and Guide

Tactix Development Project

August 8, 2026

Executive summary

The M3 ConflictAnalysis entry in the interactive proof-rule guide previously compressed learning, backjumping, assertion, and proof logging into informal placeholders. The authoritative calculus and the derived HTML guide now give the exact first-UIP construction, jump-level definition, atomic trail transition, and ordered certificate delta. This is a documentation-only clarification; solver semantics and performance are unchanged.

1 Findings and results

The old display used "derive learned clause", `backtrack(j)`, and "optional deletion" without defining the learned clause, the level j, or which deletions occur. The revised rule starts from the all-false conflict clause, resolves the latest conflict-level trail assignments with their reason clauses until one conflict-level literal remains, and identifies that literal as the negated first UIP. It defines the jump level as the maximum level among the other learned literals (or zero), and states the legal chronological target interval.

The HTML certificate is now ordered explicitly: OTFS add-before-delete pairs, the learned RUP addition with its reason replay, and eager-subsumption deletions. Root LRAT and DRAT refutations are distinguished. A new two-level worked example shows both resolution steps, the LRAT hint order, the computed jump level, and the asserted literal. The root-refutation example now shows its concrete certificate as well.

2 Evidence

The statement was checked against `src/rules/analyze/analyze.cpp`, `src/rules/backjump/backjump.cpp`, and the proof API. Documentation citation, section-reference, size, comment-hygiene, and whitespace checks pass. No solver binary was changed, so no correctness or performance claim is made from this work.

3 Risks and decisions

The guide remains a derived teaching view; `docs/calculus.md` remains authoritative. Only M3 was expanded in this increment. Several other cards still use compact quantified summaries such as "rewrites" or "resolvents"; they should be made concrete individually against their authoritative calculus and implementation rather than expanded speculatively in one broad rewrite.

4 Next actions

Review the M3 presentation in a browser. If this level of detail is preferred, apply the same notation pattern to the remaining cards with aggregate proof deltas, beginning with Substitute, Sweep, Congruence, and BVE. The solver's development next step remains the `SP-COMPACT-1` Kissat-parity row.